

Solutions — Problem Set: Directed Acyclic Graphs

BUSS975: Causal Inference in Financial Research

Solution key. The numbers in Problem 4 are produced by the code shown — the document recomputes them at render time, so the printed output and the code cannot drift apart.

For the written parts, what follows is a model answer, not the only acceptable one. Grade on whether the structure is identified correctly, not on wording.

Problem 1 (Reading a DAG)

(a) Every path from D to Y

There are exactly three. A path may not revisit a node, and U has only the single edge $U \rightarrow Y$, so no path from D to Y passes *through* U .

Path	Directed?	Middle node	Status before conditioning
$D \rightarrow M \rightarrow Y$	Yes — causal	M : chain	Open (this is the effect)
$D \leftarrow X \rightarrow Y$	No	X : fork	Open
$D \rightarrow C \leftarrow Y$	No	C : collider	Closed

Association flows along the first two; the third is blocked at C by default.

(b) Which are backdoor paths

A backdoor path must satisfy two conditions: it **connects** D and Y , and its **first edge points into** D .

- $D \leftarrow X \rightarrow Y$: connects D and Y , and begins $D \leftarrow X$, an arrow into D . **Backdoor.**
- $D \rightarrow M \rightarrow Y$: connects them, but leaves D through an outgoing arrow. Not backdoor — it is the causal path itself.
- $D \rightarrow C \leftarrow Y$: connects them, but again leaves D forward. **Not** a backdoor path, even though it is not causal either.

So there is exactly one backdoor path, $D \leftarrow X \rightarrow Y$, and closing it is the entire identification problem here.

(c) Is C a collider on every path it lies on?

In this graph, yes — but only for a reason peculiar to C : both of its edges ($D \rightarrow C$ and $Y \rightarrow C$) point into it, so C is a sink. Wherever C sits in the middle of a path, both of that path's edges at C must be the incoming ones, which is the definition of a collider.

That is an accident of C , not a general truth, and the graph contains its own counterexample. Look at Y :

- on the path $X \rightarrow Y \leftarrow U$, the two edges at Y both point in — Y is a **collider**;
- on the path $D \rightarrow M \rightarrow Y \rightarrow C$, one edge enters Y and one leaves — Y is a **chain** node.

Same variable, two paths, two roles. “Collider” describes how two particular edges meet at a node *along a given path*; it is not a label a variable carries around. This is why the question “is Z a good control?” has no answer until you say which path you mean — a point the lecture makes as *whether a variable is a good control is not a property of the variable, it is a property of its position in the graph*.

(d) Conditioning on $\{X, C\}$

X helps. It is the middle of the fork $D \leftarrow X \rightarrow Y$, the only backdoor path. Conditioning on it closes that path and removes the confounding.

C does damage, in two distinct ways.

1. C is a collider on $D \rightarrow C \leftarrow Y$, a path that was **closed**. Conditioning on it **opens** the path and manufactures an association between D and Y that does not exist in the population. Nothing else in the graph can close it again: C is the only intermediate node on that path, so the only way to block it is not to condition on C .
2. C is a **descendant of D** , so conditioning on it violates condition 2 of the backdoor criterion outright.

Verdict: the effect is *not* identified by $\{X, C\}$. It *is* identified by $\{X\}$ alone, which closes the one backdoor path, leaves the causal path $D \rightarrow M \rightarrow Y$ open, and leaves $D \rightarrow C \leftarrow Y$ closed.

Two further remarks worth making:

- Do **not** add M . It is a mediator on the causal path; conditioning on it would delete the part of the effect that travels through M , leaving only a direct effect that the question did not ask for.
- U being unobserved costs nothing. It has no arrow into D , so it lies on no backdoor path — it is a pure outcome predictor. In the lecture's terms it is a *neutral* variable: including it (if you could) would only tighten the standard error.

Problem 2 (A finance DAG: index inclusion)

(a) Every path from D to Y

Path	Type	Status
$D \rightarrow Y$	Directed — the estimand	Open
$D \leftarrow S \rightarrow Y$	Backdoor (fork at S)	Open
$D \leftarrow Q \rightarrow Y$	Backdoor (fork at Q)	Open
$D \leftarrow M \rightarrow Y$	Backdoor (fork at M)	Open
$D \rightarrow O \leftarrow Q \rightarrow Y$	Not backdoor	Closed (collider at O)

O is a sink ($D \rightarrow O$, $Q \rightarrow O$), so any path reaching O must either stop there or turn around through Q ; that yields the fifth path and no others.

(b) Why $D \rightarrow O \leftarrow Q \rightarrow Y$ is not a backdoor path

It fails the defining condition: its first edge, $D \rightarrow O$, points **out of** D , not into it. A backdoor path must begin with an arrow into D — that is what makes it a “back door.”

It is nonetheless **closed**, for a separate reason: O is a collider on it ($D \rightarrow O \leftarrow Q$), and colliders block association by default. So the path carries no correlation between D and Y — unless someone conditions on O , or on a descendant of O , which is exactly what part (d) is about.

The two facts are independent, and it is worth keeping them apart: *not a backdoor path* says you have no obligation to close it; *closed* says there is nothing to close.

(c) Is the effect identified?

If everything were observable: condition on $\{S, Q, M\}$. Each is the middle of one of the three open backdoor paths, and none is a descendant of D , so both conditions of the backdoor criterion hold. This set is also **minimal**: the three backdoor paths have no intermediate nodes in common, so dropping any one of S , Q , M leaves the corresponding path open.

Given that Q is unobserved: the achievable set is $\{S, M\}$. That blocks $D \leftarrow S \rightarrow Y$ and $D \leftarrow M \rightarrow Y$, but leaves $D \leftarrow Q \rightarrow Y$ **open**. Firm quality drives both index inclusion and investment, and no amount of conditioning on observables closes that path.

Conclusion: the effect is not identified by conditioning on observables. No regression of investment on an inclusion dummy plus size and market controls estimates the causal effect, however many controls are added — the problem is a missing variable, not an insufficient one.

What that implies for the design: abandon selection-on-observables here and find a source of variation in inclusion that is unrelated to quality. The standard route in this literature is the mechanical assignment rule — inclusion determined by a rank threshold, as in the Russell 1000/2000 reconstitution — which is a regression discontinuity, the subject of Lecture 4. Reaching that conclusion *before* running any regression is the DAG earning its keep.

(d) The referee who wants you to control for O

Reason 1: O is a collider, so controlling for it creates bias rather than removing it. Since $D \rightarrow O \leftarrow Q$, ownership is a common effect of inclusion and of unobserved quality. The path $D \rightarrow O \leftarrow Q \rightarrow Y$ is currently closed; conditioning on O opens it and induces a spurious association between D and Y running through the very variable (Q) you cannot observe. You would be adding a bias that is not there.

Reason 2: O is post-treatment, which the backdoor criterion forbids. O is a descendant of D , so including it violates condition 2 regardless of what else is true. Note also that the referee's premise — “ownership affects investment” — is an arrow $O \rightarrow Y$ that the DAG does not contain; if you granted it, O would become a **mediator** on $D \rightarrow O \rightarrow Y$, and controlling for it would strip out part of the effect you are trying to measure. On either reading of the graph, O does not belong on the right-hand side.

The general rule behind both: controls must be determined **before** treatment.

Problem 3 (Bad controls and selection)

(a) The rule, and why randomization does not save you

The rule. A bad control is a variable that is itself affected by the treatment — a **descendant of D** . Conditioning on it violates condition 2 of the backdoor criterion. Operationally: *a control must be determined before treatment*; if D could have moved it, it is not a control.

Such a variable does damage in one of two ways. If it lies on a causal path ($D \rightarrow X \rightarrow Y$) it is a **mediator**, and conditioning blocks part of the effect. If it has a second parent ($D \rightarrow X \leftarrow A \rightarrow Y$) it is a **collider**, and conditioning opens a path that was closed.

Why randomization does not protect you. Randomizing D guarantees that nothing points into D — there are no backdoor paths, so the naive comparison is already unbiased. But that is a statement about the paths that *exist*. It says nothing about paths you *create*. Conditioning on a post-treatment collider opens $D \rightarrow X \leftarrow A \rightarrow Y$, and the bias appears in a perfectly executed experiment. Randomization protects the design you were handed; it does not protect you from what you do to the data afterwards.

(b) What each sample-construction step conditions on

Study	Conditions on	Collider?
(i) funds alive at the end of the sample	Survival	Yes. Survival is a common effect of skill and luck; conditioning on it makes them correlated among survivors. The textbook case.
(ii) CRSP–Compustat merged data	Being US-listed <i>and</i> successfully matched	Yes. Both are consequences of size, age and quality — the same forces driving the outcome. Conditioned on without a line of code mentioning it.
(iii) completed deals only	Deal completion	Yes. Completion is post-announcement and a common effect of announcement terms and unobserved deal quality or market reaction.
(iv) bonds with a credit rating	Being rated	Yes, plausibly. Whether an issue is rated is a choice driven by issuer quality and by the pricing the issuer expects — both connected to the outcome.

The common thread: **selection is conditioning**. You condition on a variable whenever you include it, stratify on it, drop observations on it, or use a dataset assembled using it — and the last is the dangerous one, because it never appears in the code.

(c) A plausible bad control in each study

(i) **Disclosure mandate** → **cost of capital**. *Analyst coverage*. The mandate changes coverage, so coverage is post-treatment. If coverage is one channel through which disclosure lowers the cost of capital, it is a mediator and controlling for it removes part of the answer; if it also responds to unobserved firm quality, conditioning opens a collider path. Liquidity or bid–ask spread fail the same way.

(ii) **Bank merger** → **small-business lending**. *Post-merger branch count or local market share*. Mergers consolidate branch networks, so branches are an outcome of the merger and the main channel through which lending changes. Conditioning on them removes the mechanism, and local demand shocks — unobserved — drive both branches and lending, so a collider path opens too.

(iii) **Governance reform** → **firm value**. *Leverage, Tobin’s Q , institutional ownership, or CEO turnover* — all plausibly moved by the reform. Each is a descendant of D ; conditioning either blocks the channel or opens a path via unobserved management quality.

(d) The coefficient moves when the questionable control is added

What they should conclude. The control is doing real work: it is correlated with both D and Y , so the two specifications are estimating different things. Which one is right is a question the **DAG**

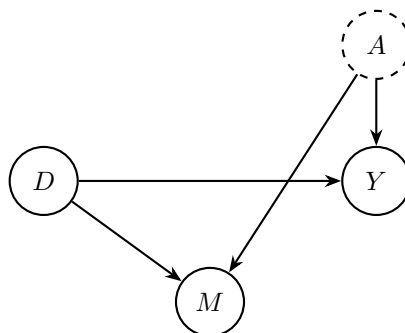
answers, not the data. If the control is post-treatment, the specification *without* it is the one that estimates the total effect, and the movement is a measurement of bad-control bias — a diagnostic, and worth reporting as such.

What they should not conclude. Not that the smaller estimate is “more conservative,” nor that more controls are safer, nor that the movement itself adjudicates between the specifications. A coefficient that moves is not a specification test: it tells you the control matters, never that including it is correct. And the converse fails too — a coefficient that barely moves is not evidence of no confounding, since an omitted variable can be strong and the included control weak.

The honest report is both specifications, plus a sentence saying which one the assumed graph supports and why.

Problem 4 (Simulation: a bad control with randomized treatment)

(a) The DAG, and what to expect



$M = D + A + e_M$ gives $D \rightarrow M$ and $A \rightarrow M$; $Y = \beta D + \gamma A + e_Y$ gives $D \rightarrow Y$ and $A \rightarrow Y$. There is **no arrow** $M \rightarrow Y$.

M is a collider, and only a collider. It is a common effect of D and A . It is *not* a mediator: nothing travels from D to Y through M , because M does not enter the equation for Y .

Backdoor paths: none. Nothing points into D — it is drawn independently, which is what “randomly assigned” means here. The only non-causal path, $D \rightarrow M \leftarrow A \rightarrow Y$, leaves D forward (so it is not a backdoor path) and is closed anyway (collider at M).

So the regression of Y on D alone should return $\beta = 2$. With no open backdoor path there is nothing to adjust for; A is omitted, but A is independent of D , so it inflates the residual variance without biasing the slope.

(b) The three regressions

```

set.seed(975)
n <- 200000; beta <- 2; gamma <- 4; sM2 <- 1
D <- rnorm(n); A <- rnorm(n)
M <- D + A + rnorm(n, 0, sqrt(sM2))
Y <- beta * D + gamma * A + rnorm(n)
c(Y_on_D      = coef(lm(Y ~ D))["D"]),
  Y_on_D_M     = coef(lm(Y ~ D + M))["D"]),
  Y_on_D_M_A   = coef(lm(Y ~ D + M + A))["D"]))

```

```

      Y_on_D      Y_on_D_M      Y_on_D_M_A
1.999064645 -0.003641445  2.001581749

```

Y on D returns 2.00 — correct, as predicted. Y on D and M returns 0.00. Y on D , M and A returns 2.00 again.

A researcher who ran only the second regression would conclude that D **has no effect on** Y — and would feel well justified, having “controlled for” a variable strongly correlated with both. They are wrong because M is a collider: conditioning on it opens $D \rightarrow M \leftarrow A \rightarrow Y$, and the association that path transmits happens to be equal and opposite to the true effect. The third regression restores the truth by conditioning on A , which closes the newly opened path — but A is unobserved in any real study, so that repair is not available in practice.

Note how bad this failure mode is. The treatment was randomly assigned, the control is a real variable measured without error, and the estimate is not merely biased but exactly zero.

(c) Derivation

All variables are mean zero, so the normal equations for $Y = b_D D + b_M M$ are

$$\begin{aligned} b_D \text{var}(D) + b_M \text{cov}(D, M) &= \text{cov}(D, Y), \\ b_D \text{cov}(D, M) + b_M \text{var}(M) &= \text{cov}(M, Y). \end{aligned}$$

With D, A, e_M, e_Y mutually independent and $\text{var}(D) = \text{var}(A) = 1$:

$$\begin{aligned} \text{var}(D) &= 1, & \text{cov}(D, M) &= \text{cov}(D, D + A + e_M) = 1, \\ \text{var}(M) &= 1 + 1 + \sigma_M^2 = 2 + \sigma_M^2, & \text{cov}(D, Y) &= \text{cov}(D, \beta D + \gamma A + e_Y) = \beta, \\ \text{cov}(M, Y) &= \beta \text{var}(D) + \gamma \text{var}(A) = \beta + \gamma. \end{aligned}$$

Substituting:

$$b_D + b_M = \beta, \quad b_D + b_M (2 + \sigma_M^2) = \beta + \gamma.$$

Subtracting the first from the second leaves $b_M (1 + \sigma_M^2) = \gamma$, so

$$b_M = \frac{\gamma}{1 + \sigma_M^2}, \quad b_D = \beta - b_M = \beta - \frac{\gamma}{1 + \sigma_M^2}. \quad \blacksquare$$

At $\beta = 2, \gamma = 4, \sigma_M^2 = 1$: $b_D = 2 - 4/2 = 0$, which is why the coefficient vanished exactly.

```

check <- function(sM2, n = 200000, beta = 2, gamma = 4) {
  set.seed(975)
  D <- rnorm(n); A <- rnorm(n)
  M <- D + A + rnorm(n, 0, sqrt(sM2))
  Y <- beta * D + gamma * A + rnorm(n)
  c(sigma_M2 = sM2,
    simulated = coef(lm(Y ~ D + M))["D"],
    formula   = beta - gamma / (1 + sM2))
}
round(rbind(check(0.25), check(1), check(4)), 4)

```

	sigma_M2	simulated	formula
[1,]	0.25	-1.2017	-1.2
[2,]	1.00	-0.0036	0.0
[3,]	4.00	1.1974	1.2

The formula matches the simulation at every value: -1.20 , 0.00 and $+1.20$.

(d) Why the bias shrinks as M gets noisier

In words. The noisier the collider, the smaller the damage from conditioning on it. As $\sigma_M^2 \rightarrow \infty$ the bias term $\gamma/(1 + \sigma_M^2) \rightarrow 0$ and $b_D \rightarrow \beta$: a sufficiently noisy M is harmless. At the other end, $\sigma_M^2 \rightarrow 0$ makes M a perfect record of $D + A$ and the bias is at its worst, $b_D \rightarrow \beta - \gamma = -2$.

Why it feels wrong. Every instinct trained on measurement error says noise is bad: a noisy regressor is attenuated, a noisy instrument is weak, a noisy outcome is imprecise. Here noise *helps*.

Why it is right. The two cases are not the same, because M is not being used to measure anything. The harm from conditioning on a collider comes entirely from what M tells you about the **unobserved** A . Knowing M and D pins down $A + e_M$; how much that reveals about A depends on the signal-to-noise ratio $1 : \sigma_M^2$. Add noise and each value of M is consistent with a wider range of A , so conditioning on it induces less spurious D – A correlation — and in the limit, where M is pure noise, conditioning on it is conditioning on nothing at all.

The practical lesson runs the other way from the intuition: **the cleaner and better-measured a post-treatment control is, the more damage it does**. A precisely measured bad control is more dangerous than a sloppy one.